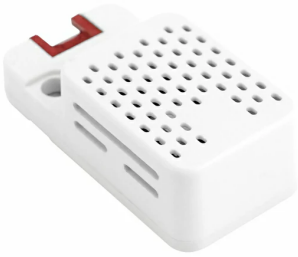
Unit CO2

SKU:U103



















Description

Unit CO2 is a digital air CO_2 concentration detection unit, built-in with Sensirion's **SCD40** sensor and power step-down circuit, using I2C communication. This unit is suitable for air environment condition measurement, with a typical accuracy of ± (50 ppm + 5% of reading) for CO_2 measurement, and a range of 400 ppm ~ 2000 ppm, while also measuring ambient temperature and humidity.

Features

- CO2 concentration detection
- Temperature and humidity measurement
- I2C communication (0x62)
- Integrated 5V -> 3.3V DC-DC
- 2 x LEGO compatible holes
- HY2.0 4P interface

Includes

- 1 x Unit CO2
- 1 x HY2.0 connection cable (20CM)

Applications

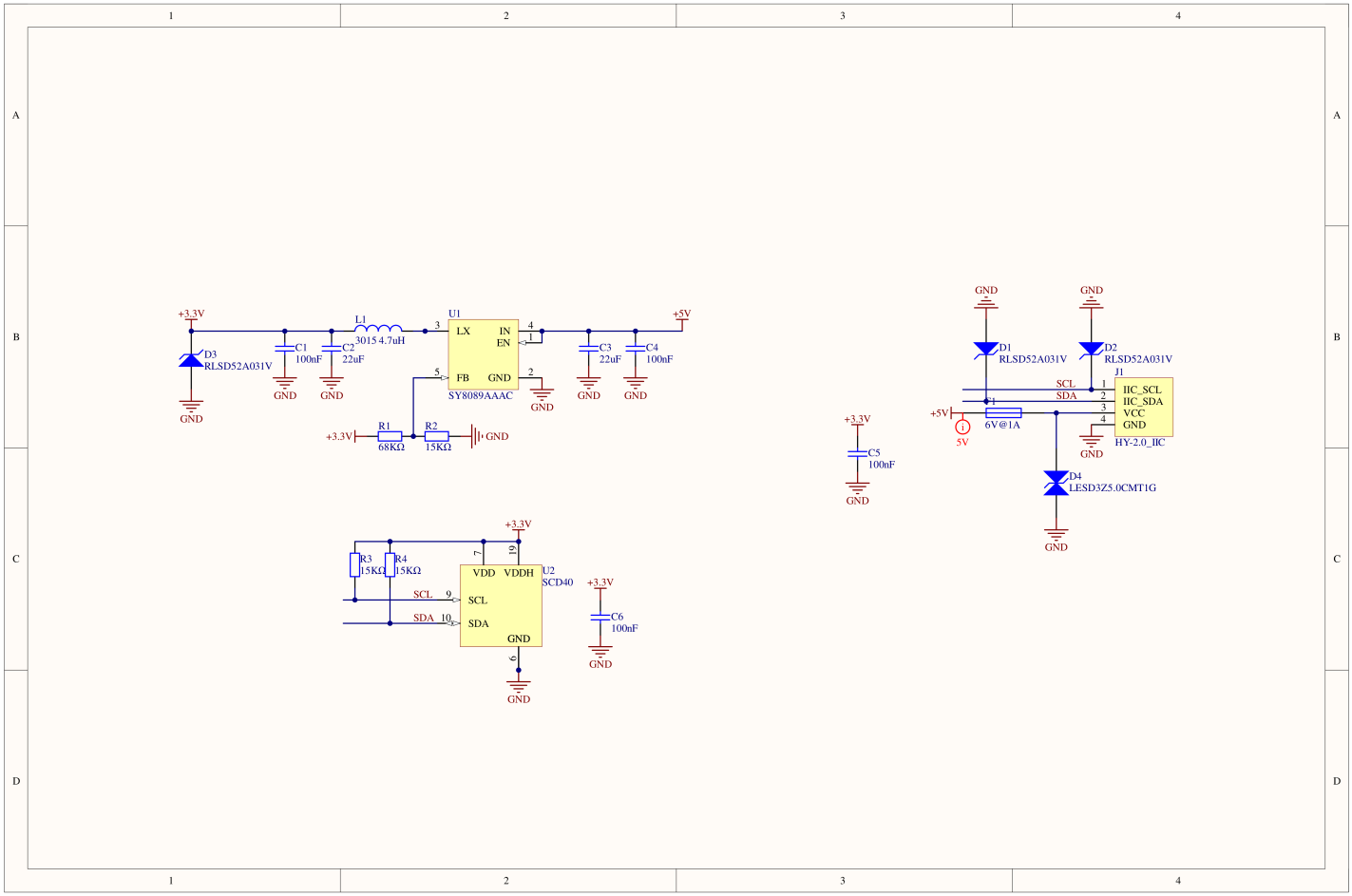
- CO2 concentration detection
- Temperature and humidity monitoring

Specifications

Specification	Parameter
CO2 Measurement Range	400 ~ 2000 ppm
CO2 Sampling Accuracy	± (50 ppm + 5% of reading)
Temperature Range	-10 ~ 60 °C
Humidity Range	0 ~ 95 %RH
Communication Protocol	I2C:0x62
Enclosure Material	Plastic (PC)
Product Size	48.0 x 24.0 x 16.0mm
Product Weight	7.6g
Package Size	138.0 x 93.0 x 17.0mm
Gross Weight	13.0g

Schematics

- Unit CO2 Schematics PDF

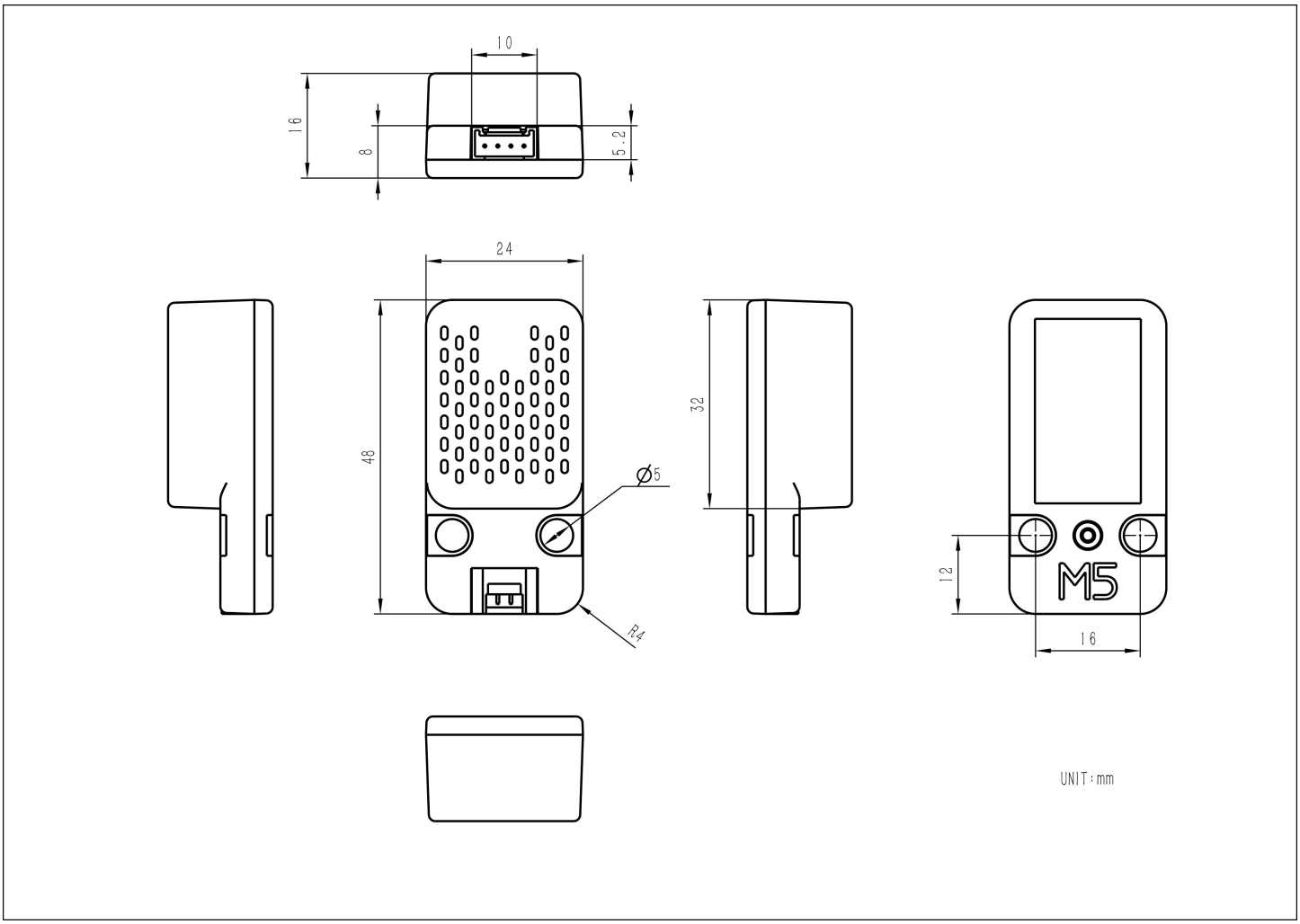


PinMap

Unit CO2

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Model Size



Structure

- Unit CO2 Structure Files

Datasheets

- SCD40
- SY8089AAAC

Softwares

Arduino

- Unit CO2 Arduino Tutorial
- Unit CO2 with M5Atom
- Unit CO2 with M5AtomS3
- Unit CO2 with M5AtomS3Lite
- Unit CO2 with M5Core
- Unit CO2 with M5Core2
- Unit CO2 with M5StickC
- Unit CO2 with M5StickCPlus

UiFlow1

- Unit CO2 UiFlow1 Docs

UiFlow2

- Unit Co2 UiFlow2 Docs

Home Assistant

- Unit CO2 Home Assistant Integration